

AI Usage Note

Project: Disk Cleanup AI – Discord Bot Integration

AI tools were used as development assistants throughout the implementation of the Disk Cleanup AI project. Their primary role was to support architecture planning, code generation, debugging, and documentation while all final decisions, testing, and validation were performed manually.

What AI Helped With

AI assisted in designing the integration between the existing Discord bot and the LLM-based cleanup pipeline consisting of the scanner, classifier, and cleanup modules. It helped identify the mismatch between the `FileEntry` objects used by the bot and the dictionary outputs returned by the classifier and suggested an adapter-based approach without requiring major refactoring.

AI also generated implementation scaffolding and code suggestions across multiple modules, including `models.py`, `scanner.py`, `classifier.py`, `discord_bot.py`, `cleanup.py`, and `main.py`. These suggestions were reviewed and adapted before being incorporated into the project.

For feature development, AI helped design the duplicate file detection workflow using size grouping and SHA-256 hashing, and assisted in documenting project functionality and maintaining consistency between implementation and documentation.

What AI Got Wrong

Some AI-generated suggestions required correction. Initial integration guidance did not account for differences in cleanup function signatures, requiring additional iterations after the actual code was shared. Configuration naming mismatches and package import issues also required manual debugging. In addition, AI occasionally introduced redundant code that was removed during review.

These cases highlighted the importance of verifying AI-generated outputs rather than accepting them directly.

Most Effective Prompting Practices

The most useful responses were obtained by sharing the actual source code instead of describing the problem abstractly. Providing complete error messages and current folder structures significantly improved debugging accuracy. Constraint-based prompts such as requesting solutions without changing the project structure also produced more practical recommendations.

AI Tools Used

- Claude
- ChatGPT
- Ollama
- Qwen2.5:3B

Conclusion

AI served as a development assistant rather than an autonomous author. All AI-generated suggestions were treated as starting points, and every modification was manually reviewed, tested, and validated before being incorporated into the final project.